Chapter 112

Optical Flow

This chapter details the Optical Flow nodes available in Fusion.

The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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The OpticalFlow node

Optical Flow Node Introduction

This node analyzes a clip connected to its input using an Optical Flow algorithm. Think of optical flow as a per-pixel motion vector that matches up features over several frames.

The computed optical flow is stored within the Vector and Back Vector aux channels of the output. These channels can be used in other nodes like the Vector Motion Blur or Vector Distort. However, Optical Flow must render twice when connecting it to a Time Stretcher or Time Speed node. These nodes require the channels A. FwdVec and B. BackVec in that order, but Optical Flow generates A. BackVec and A. FwdVec when it processes.

If you find that optical flow is too slow, consider rendering it out into OpenEXR files using a Saver node.

TIP: If the footage input flickers on a frame-by-frame basis, it is a good idea to deflicker the footage beforehand.

Inputs

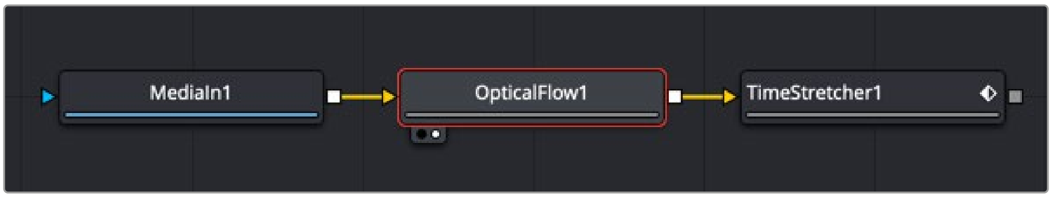
The Optical Flow node includes a single orange image input.

Input: The orange background input accepts a 2D image. This is the sequence of frames for which you want to compute optical flow. The output of the Optical Flow node includes the image and vector channels. The vector channels can be displayed by right-clicking in the viewer and choosing Channel > Vectors and then Options > Normalize Color Range.

Basic Node Setup

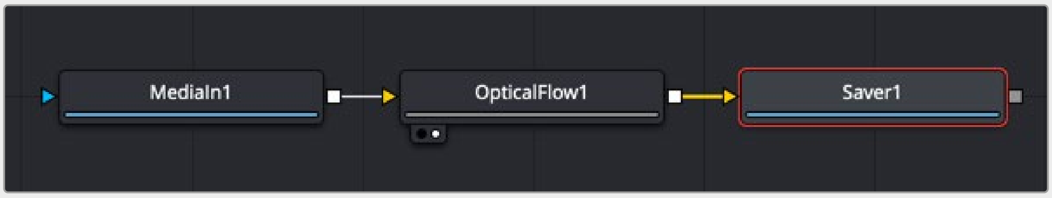
The Optical Flow node analyzes the frames connected to the background input. Trimming a Loader or MediaIn to only the range you need prevents analyzing unnecessary frames. The output of the node can then be connected to any node that takes advantage of vector channels, such as a Time Stretcher.

TIP: When analyzing Optical Flow vectors, consider adding a Smooth Motion node afterward with smoothing for forward/ backward vectors enabled.



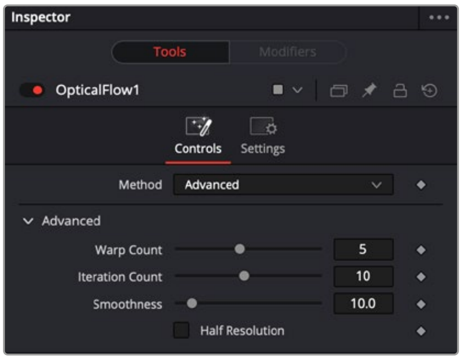
An Optical Flow node generating motion vectors on-the-fly.

Alternatively, if you find the Optical Flow node too slow to analyze the frames, consider rendering it out to an OpenEXR format using a Saver node. Then import the rendered EXR file as your new image with embedded vector channels.



An Optical Flow node rendered out through a Saver node.

Inspector



Optical Flow advanced controls

Controls Tab (Advanced)

When you add an Optical Flow, Repair Frame, or Tween node to a Comp, a Method drop-down menu in the Inspector allows you to choose between an Advanced GPU-based algorithm or a Classic CPU-based algorithm. This Advanced method is the same Optical Flow algorithm used in other DaVinci Resolve pages.

Warp Count

Decreasing this slider makes the optical flow computations faster. To understand what this option does, you must understand that the optical flow algorithm progressively warps one image until it matches with the other image. After some point, convergence is reached, and additional warps become a waste of computational time. You can tweak this value to speed up the computations, but it is good to watch what the optical flow is doing at the same time.

Iteration Count

Decreasing this slider makes the computations faster. In particular, just like adjusting the Warp Count, adjusting this option higher will eventually yield diminishing returns and not produce significantly better results. By default, this value is set to something that should converge for all possible shots and can be tweaked lower fairly often without reducing the disparity's quality.

Smoothness

This controls the smoothness of the optical flow. Higher smoothness helps deal with noise, while lower smoothness brings out more detail.

Half Resolution

The Half Resolution checkbox is used purely to speed up the calculation of the optical flow. The input images are resized down and tracked to produce the optical flow.

Controls Tab (Classic)

By choosing Classic from the Method drop-down menu in the Inspector, you can use the older CPU-based algorithm to maintain compatibility with Comps created in previous versions. This method may also be better suited for some Stereo3D processing.

When using the Classic method, a single slider at the top of the Inspector improves performance by generating proxies. The remaining Advanced section parameters tune the Optical Flow vector calculations. The default settings serve as a good standard. In most cases, tweaking of the advanced settings is not needed. Many deliver small or diminishing returns. However, depending on the settings, rendering time can easily vary by 10x. If you're interested in reducing process time, it is best to start by experimenting with the Proxy, Number of Iterations, and Number of Warps sliders and changing the filtering to Bilinear.

Proxy (for Tracking)

The Proxy slider is used purely to speed up the calculation of the optical flow. The input images are resized down by the proxy scale and tracked to produce the optical flow. The computational time is roughly proportional to the number of pixels in the image. This means a proxy scale of 2 will give a 4x speedup, and a proxy scale of 3 will give a 9x speedup.

Smoothness

This controls the smoothness of the optical flow. Higher smoothness helps deal with noise, while lower smoothness brings out more detail.

Edges

This slider is another control for smoothness but applies it based on the color channel. It tends to have the effect of determining how edges in the flow follow edges in the color images. When it is set to a low value, the optical flow becomes smoother and tends to overshoot edges. When it is set to a high value, details from the color images start to slip into the optical flow, which is not desirable. Edges in the flow end up more tightly aligning with the edges in the color images. This can result in streaked-out edges when the optical flow is used for interpolation. As a rough guideline, if you are using the disparity to produce a Z-channel for post effects like Depth of Field, then set it lower in value. If you are using the disparity to perform interpolation, you might want it to be higher in value.

Match Weight

This control sets a threshold for how neighboring groups of foreground/background pixels are matched over several frames. When set to a low value, large structural color features are matched. When set to higher values, small sharp variations in the color are matched. Typically, a good value for this slider is in the [0.7, 0.9] range. When dealing with stereo 3D, setting this option higher tends to improve the matching results in the presence of differences due to smoothly varying shadows or local lighting variations between the left and right images. The user should still perform a color match or deflickering on the initial images, if necessary, so they are as similar as possible. This option also helps with local variations like lighting differences due to light passing through a mirror rig.

Mismatch Penalty

This option controls how the penalty for mismatched regions grows as they become more dissimilar. The slider provides a choice between a balance of Quadratic and Linear penalties. Quadratic strongly penalizes large dissimilarities, while Linear is more robust to dissimilar matches. Moving this slider toward Quadratic tends to give a disparity with more small random variations in it, while Linear produces smoother, more visually pleasing results.

Warp Count

Decreasing this slider makes the optical flow computations faster. In particular, the computational time depends linearly upon this option. To understand what this option does, you must understand that the optical flow algorithm progressively warps one image until it matches with the other image. After some point, convergence is reached, and additional warps become a waste of computational time. The default value in Fusion is set high enough that convergence should always be reached. You can tweak this value to speed up the computations, but it is good to watch what the optical flow is doing at the same time.

Iteration Count

Decreasing this slider makes the computations faster. In particular, the computational time depends linearly upon this option. Just like adjusting the Warp Count, adjusting this option higher will eventually yield diminishing returns and not produce significantly better results. By default, this value is set to something that should converge for all possible shots and can be tweaked lower fairly often without reducing the disparity's quality.

Filtering

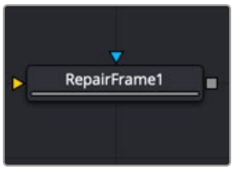
This option controls filtering operations used during flow generation. Catmull-Rom filtering will produce better results, but at the same time, turning on Catmull-Rom will increase the computation time steeply.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Optical Flow nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Repair Frame [REP]



The Repair Frame node

Repair Frame Node Introduction

Repair Frame replaces damaged or missing frames or portions of frames with scratches or other temporally transient artifacts. It requires three frames: the repair frame and two neighboring frames. An Optical Flow node is not required for generating motion vectors since the Repair Frame node computes the optical flow. However, this can make it slow to process.

Repair Frame will not pass through, but rather destroys, any aux channels after the computation is done.

See the Optical Flow node for controls and settings information.

TIP: If your footage varies in color from frame to frame, sometimes the repair can be noticeable because, to fill in the hole, Repair Frame must pull color values from adjacent frames. Consider using deflickering, color correction, or using a soft-edged mask to help reduce these kinds of artifacts.

Inputs

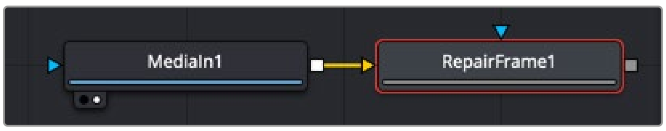
There are two inputs on the Repair Frame node. One is used to connect a 2D image that will be repaired and the other is for an effect mask.

Input: The orange input is used for the primary 2D image that will be repaired.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the repairs to certain areas.

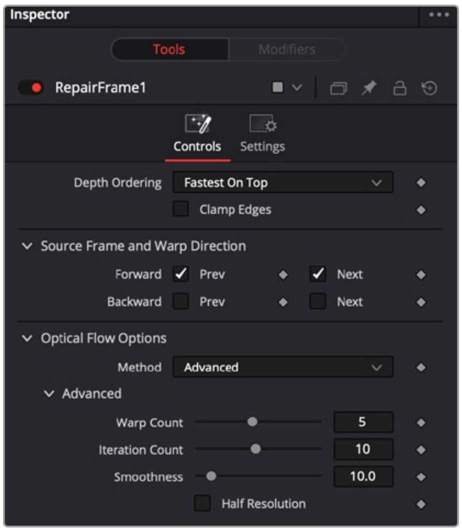
Basic Node Setup

The Repair Frame node analyzes the incoming MediaIn node and repairs single frame issues like dust or scratches.



A Repair Frame node set up to analyze a MediaIn node using internal optical flow analysis.

Inspector



The Repair Frame Controls tab

Controls Tab

The Controls tab includes options for how to repair the frames. It also includes controls for adjusting the optical flow analysis, identical to those controls in the Optical Flow node.

Depth Ordering

The Depth Ordering determines which parts of the image should be rendered on top by selecting either Fastest On Top or Slowest On Top. The examples below best explain these options.

In a locked-off camera shot where a car is moving through the frame, the background does not move, so it produces small, or slow, vectors, while the car produces larger, or faster, vectors.

The depth ordering in this case is Fastest On Top since the car draws over the background.

In a shot where the camera pans to follow the car, the background has faster vectors, and the car has slower vectors, so the Depth Ordering method is Slowest On Top.

Clamp Edges

Under certain circumstances, this option can remove the transparent gaps that may appear on the edges of interpolated frames. Clamp Edges causes a stretching artifact near the edges of the frame that is especially visible with objects moving through it or when the camera is moving.

Because of these artifacts, it is a good idea to use clamp edges only to correct small gaps around the edges of an interpolated frame.

Edge Softness

This slider is displayed only when Clamp Edges is enabled. The slider helps to reduce the stretchy artifacts that might be introduced by Clamp Edges.

If you have more than one of the Source Frame and Warp Direction checkboxes turned on, this can lead to doubling up of the stretching effect near the edges. In this case, you'll want to keep the softness rather small at around 0.01. If you have only one checkbox enabled, you can use a larger softness at around 0.03.

Source Frame and Warp Direction

These checkboxes allow you to choose which frames and vectors create the in-between frames. Each method ticked on will be blended into the result.

- **Prev Forward:** Takes the previous frame and uses the Forward vector to interpolate the new frame.
- **Next Forward:** Takes the next frame in the sequence and uses the Forward vector to interpolate the new frame.
- **Prev Backward:** Takes the previous frame and uses the Back Forward vector to interpolate the new frame.
- **Next Backward:** Takes the next frame in the sequence and uses the Back vector to interpolate the new frame.

Optical Flow Options

These settings tweak the optical flow analysis. See the Classic and Advanced Controls section for the Optical Flow node earlier in this chapter.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Optical Flow nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Smooth Motion [SM]

The Smooth Motion node

Smooth Motion Node Introduction

The Smooth Motion node smooths various AOV (Arbitrary Output Variables) channels in a clip using optical flow to look at neighboring frames. It can be used for smoothing the Disparity channel in a stereo 3D clip, where it helps reduce temporal edge/fringing artifacts, but it can also smooth a wide range of channels like vectors, normals, and Z.

It is required that the image connected to the input on the node have precomputed Vector and Back Vector channels; otherwise, this tool prints error messages in the Console window.

Check on the channels you want to temporally smooth. Be aware that if a channel selected for smoothing is not present, Smooth Motion will not fail, nor will it print any error messages.

It can also be used to smooth the Vector and Back Vector channels; however, sometimes, this can make the interpolated results worse if there are conflicting motions or objects in the shot that move around erratically, jitter, or bounce rapidly.

TIP: You can use two or more Smooth Motion nodes in sequence to get additional smoothing. With one Smooth Motion node, the previous, current, and next frames are examined for a total of 3; with two Smooth Motion nodes, 5 frames are examined; and with three Smooth Motion nodes, 7 frames are examined.

Another technique using two Smooth Motion nodes is to use the first Smooth Motion node to smooth the Vector and Back Vector channels. Use the second Smooth Motion to smooth the channels you want to smooth (e.g., Disparity). This way, you use the smoothed vector channels to smooth Disparity.

You can also try using the smoothed motion channels to smooth the motion channels.

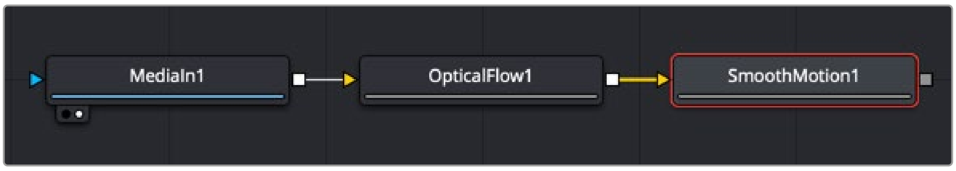
Inputs

The Smooth Motion node includes a single orange image input.

Input: The orange image input accepts a 2D image. This is the sequence of images for which you want to compute smooth motion. This image must have precomputed Vector and Back Vector channels either generated from an Optical Flow node or saved in EXR format with vector channels.

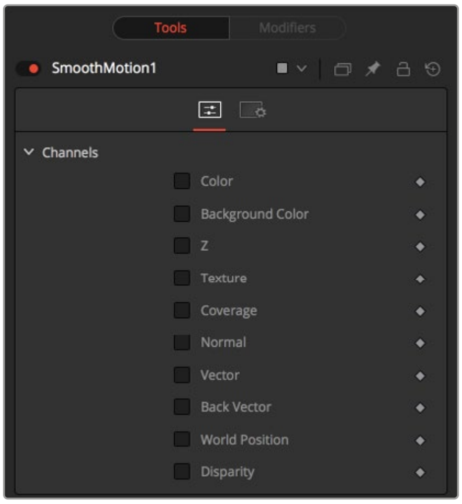
Basic Node Setup

The Smooth Motion node takes the output of the Optical Flow node for the required Vector and Back Vector channels. The Smooth Motion node can then be used to smooth those channels or AO channels.



A Smooth Motion node using Vector and Back Vector channels from the Optical Flow node.

Inspector



The Smooth Motion Controls tab

Controls Tab

The Controls tab includes checkboxes for the channels you want to smooth. If a channel selected for smoothing is not available in the input image, Smooth Motion will not fail, nor will it print any error messages to the Console.

Channel

Smooth Motion can be applied to more than just the RGBA channels. It can also be applied to the other AOV channels.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Optical Flow nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Tween [Tw]

The Tween node

Tween Node Introduction

Tween reconstructs a missing frame by interpolating between two neighboring frames using the optical flow. Tween is nearly identical in functionality to Time Speed and Time Stretcher. The major difference is that it works on two images that are not serial members of a sequence. As a consequence, it cannot use the Vector or Back Vector aux channels stored in the images. The Tween node manually generates the optical flow, so there is no need to add an Optical Flow node before the Tween node. The generated optical flow is thrown away and is not stored back into the output frames.

Since optical flow is based on color matching, it is a good idea to color correct your images to match ahead of time. Also, if you are having trouble with noisy images, it may also help to remove some of the noise ahead of time.

Tween destroys any input aux channels. See the Optical Flow node for controls and settings information.

Inputs

There are two image inputs on the Tween node and an effects mask input.

Input 0: The orange input, labeled input 0, is the previous frame to the one you are generating.

Input 1: The green input, labeled input 1, is the next frame after the one you are generating.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the Tween to certain areas.

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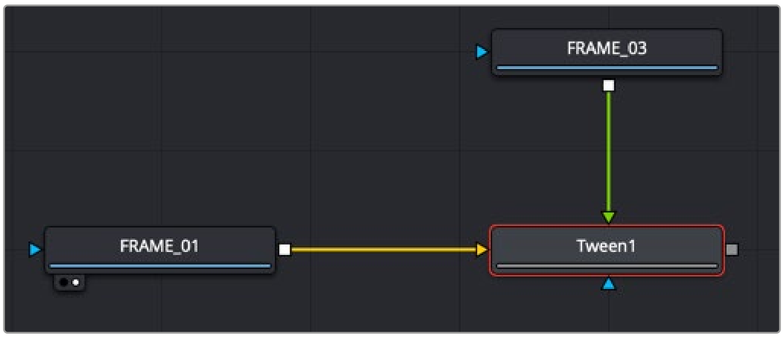
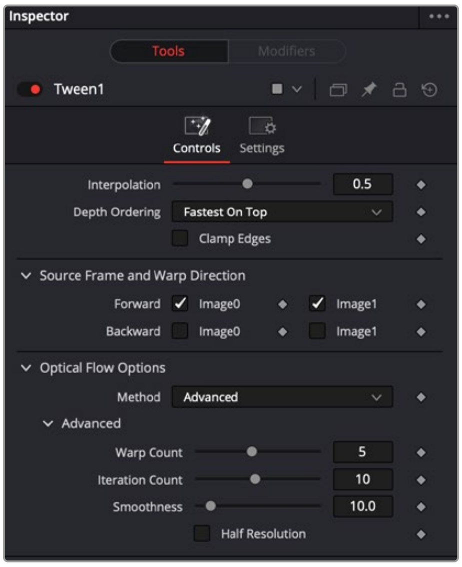
Basic Node Setup

The Tween node receives two inputs for the two neighboring frames to the one you are generating. Below, the previous frame, frame 01, is connected to the orange input 0. The next frame, frame 03, is connected to the green input 1. The Tween node will generate frame 02 and output the sequence.



The Tween node receives two neighboring frames and generates the middle one.

Inspector



The Tween Controls tab

Controls Tab

The Controls tab includes options for how to tween frames. It also includes controls for adjusting the optical flow analysis, identical to those controls in the Optical Flow node.

Interpolation Parameter

This option determines where the frame you are interpolating is, relative to the two source frames A and B. An Interpolation Parameter of 0.0 will result in frame A, a parameter of 1.0 will result in frame B, and a parameter of 0.5 will yield a result halfway between A and B.

Depth Ordering

The Depth Ordering determines which parts of the image should be rendered on top by selecting either Fastest On Top or Slowest On Top. The examples below best explain these options.

In a locked-off camera shot where a car is moving through the frame, the background does not move, so it produces small, or slow, vectors, while the car produces larger, or faster, vectors.

The Depth Ordering in this case is Fastest On Top since the car draws over the background.

In a shot where the camera pans to follow the car, the background has faster vectors, and the car has slower vectors, so the Depth Ordering method is Slowest On Top.

Clamp Edges

Under certain circumstances, this option can remove the transparent gaps that may appear on the edges of interpolated frames. Clamp Edges causes a stretching artifact near the edges of the frame that is especially visible with objects moving through it or when the camera is moving.

Because of these artifacts, it is a good idea to use Clamp Edges only to correct small gaps around the edges of an interpolated frame.

Edge Softness

This slider is displayed only when Clamp Edges is enabled. The slider helps to reduce the stretchy artifacts that might be introduced by Clamp Edges.

If you have more than one of the Source Frame and Warp Direction checkboxes turned on, this can lead to doubling up of the stretching effect near the edges. In this case, you'll want to keep the softness rather small at around 0.01. If you have only one checkbox enabled, you can use a larger softness at around 0.03.

Source Frame and Warp Direction

These checkboxes allow you to choose which frames and vectors create the in-between frames. Each method ticked on will be blended into the result.

- **Prev Forward:** Takes the previous frame and uses the Forward vector to interpolate the new frame.
- **Next Forward:** Takes the next frame in the sequence and uses the Forward vector to interpolate the new frame.
- **Prev Backward:** Takes the previous frame and uses the Back Forward vector to interpolate the new frame.
- **Next Backward:** Takes the next frame in the sequence and uses the Back vector to interpolate the new frame.

Optical Flow Options

These settings tweak the optical flow analysis. See the Class and Advanced Controls section for the Optical Flow node earlier in this chapter.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Optical Flow nodes. These common controls are described in detail in the following "The Common Controls" section.

Vector Warping Toolset
(Studio Version Only)

You can use the Vector Warping 2D imaging toolset below to map and warp a reference frame to all the other frames in the sequence. This is useful in face replacements, digital makeup, sign replacements, etc. to track and warp assets in tandem with the background. At its simplest, you:

- Apply VFX changes—make-up, face paint, scars, black eyes—on one frame,
- Ensure the areas of the reference frame are visible in the frames you are tracking,
- Add Vector Warp, set reference frame, select Generate + Texture Map and track the changes to the sequence using optical flow.

Vector Denoise [VDn]

The Vector Denoise node

Vector Denoise Node Introduction

This tool allows for a general purpose averaging operation across neighboring frames. While similar to temporal noise reduction, the source pixels to average are motion vector compensated.

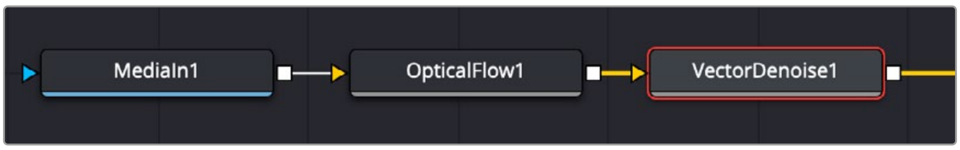
Inputs

The Vector Denoise node includes a single orange image input.

Input: The orange image input accepts a 2D image. This image must have precomputed Vector and Back Vector channels, either generated from an Optical Flow node or saved in EXR format with vector channels.

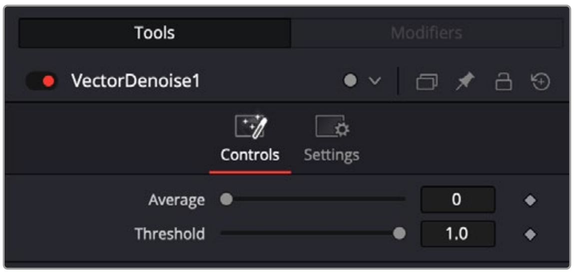
Basic Node Setup

The Vector Denoise node takes the output of the Optical Flow node for the required Vector and Back Vector channels and removes noise across multiple frames.



A Vector Denoise node using Vector and Back Vector channels from the Optical Flow node

Inspector



The Vector Denoise Controls tab

Controls Tab

The Controls tab lets you adjust the Vector Denoise settings.

- **Average:** Define the temporal window in number of frames.
- **Threshold:** Set an upper threshold to ignore flashes and brief highlights when averaging pixel values.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Optical Flow nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Vector Transform [VXf]

The Vector Transform node

Vector Transform Node Introduction

This tool allows custom transforms on the vector channel information, allowing tracking information to be resized, rotated and translated to be applied on another image or sequence. Use this to simulate transfer of kinetic energy, object responses and difference in material properties in a dynamic scene.

Inputs

The Vector Transform node includes an orange image input, and an attenuate mask input.

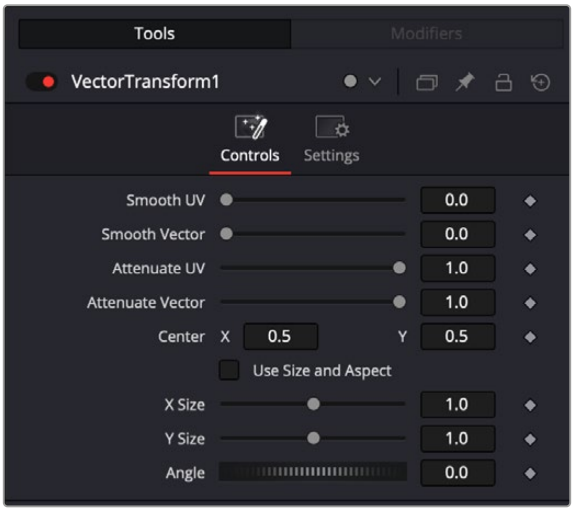
- **Input:** The orange image input accepts a 2D image.
- **Attenuate Mask:** The white input accepts a mask shape.

Basic Node Setup

The Vector Transform node takes the output of the Optical Flow node for the required Vector and Back Vector channels and transforms vector channel information.

A Vector Transform node using Vector and Back Vector channels from the Optical Flow node

Inspector



The Vector Transform Controls tab

Controls Tab

The Controls tab lets you adjust the Vector Transform settings.

- **Smooth UV:** Use this to average the results with the neighboring UV values. This can be used to smooth out anomalies and unwanted variations in the UV channels.
- **Smooth Vector:** Use this to smoothen vectors across neighboring frames. Useful as a pre-processing step before a Vector Warp with texture map.
- **Attenuate UV:** Amplify, decrease, or zero out the warp map effect from the vector magnitudes. This control can also decrease the strength of the UV channels.
- **Attenuate Vector:** Reduces the motion vector length, letting you control the amount of warping applied from full down to nothing.
- **Center X/Y:** Use this control to reposition the UV channels.
- **Size X/Y:** Use this control to rescale the UVs.
- **Use Size and Aspect:** Check this box to activate the parameters below.

  Size: Scales the image equally along both the X and Y axes.

  Aspect: Lets you set the aspect ratio of the image.

- **Angle:** Adjusts the angle of the UV.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Optical Flow nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Vector Warp [VWp]

The Vector Warp node

Vector Warp Node Introduction

The Vector Warp node has a main image input sequence (the background) and the map input image (a still to be warped). The output is an RGBA image and may contain extra motion vector channels (Vx, Vy. BVx, BVy) with forward and backward tracking information for use in downstream tools.

This tool distorts a nominated reference frame or image according to the motion of the source clip. The tool has two inputs: the Input Layer, which is for your source clip along with its motion vectors (generated using an Optical Flow tool) and the Texture Layer. Any clip or image connected to the Texture Layer will be warped based on the movement of the incoming vectors from the Input Layer. For warping to occur, motion vectors need to be generated for the source clip.

It is generally expected that Vector Warp follows an Optical Flow tool, which creates the required vectors.

Because Vector Warp uses motion vectors to follow movement in a clip, it's particularly useful for adding or removing elements from non-solid surfaces like fabric or skin. This tool can be used to add or remove logos from clothing, add a scar to someone's face, or add texture to skin

Inputs

The Vector Warp node uses two inputs.

Input: The orange image input accepts a 2D image sequence. This image must have precomputed Vector and Back Vector channels, either generated from an Optical Flow node or saved in EXR format with vector channels.

Texture: The green input accepts a 2D still image to be used as the texture for warping.

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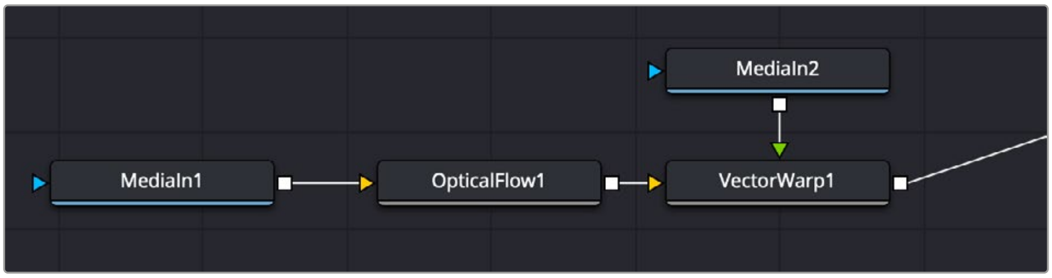
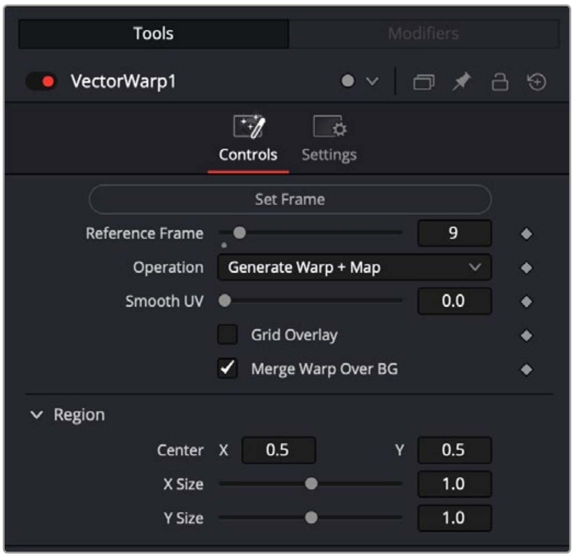
Basic Node Setup

The Vector Warp node takes the output of a clip and a 2D still image and warps them together.



A Vector Warp node

Inspector



The Vector Warp Controls tab

Controls Tab

The Controls tab lets you adjust the Vector Warp settings.

- **Set Frame:** Uses the current frame as the reference frame. Pressing this button sets the Reference Frame below.
- **Reference Frame:** Set the reference frame for the Main sequence that the Map overlay was created for.
- **Operation:** Choose the operation you wish to perform.

  Generate Warp: Creates a warp map from the reference frame to the current frame, saved in the UV channels.

  Generate Warp + Map: Creates a warp map from the reference frame to the current frame and warps the map frame.

  Apply Warp Map: Warps the map frame from a previously generated texture map from input.

  UnWarp: Freezes the map at reference frame and reverse warps the main current frame to resemble the reference frame.

- **Smooth UV:** This will average the results with the neighboring UV values. This can be used to smooth out anomalies and unwanted variations in the UV channels.
- **Grid Overlay:** Displays the warping as a viewer overlay.
- **Merge Warp over BG:** Displays the composited map image on the main image sequence.

Common Controls

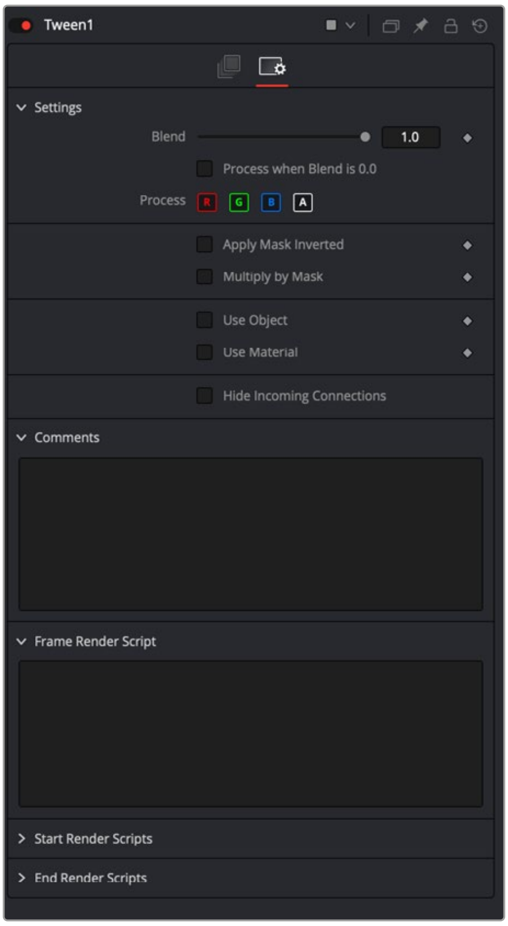
Settings Tab

The Settings tab in the Inspector is also duplicated in other Optical Flow nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

The Common Controls

Nodes that handle optical flow operations share a number of identical controls in the Inspector. This section describes controls that are common among Optical Flow nodes.

Inspector



The Common Optical Flow Settings tab

Settings Tab

The Settings tab in the Inspector can be found on every tool in the Optical Flow category. The controls are consistent and work the same way for each tool.

Blend

The Blend control is used to blend between the tool's original image input and the tool's final modified output image. When the blend value is 0.0, the outgoing image is identical to the incoming image. Normally, this causes the tool to skip processing entirely, copying the input straight to the output.

Process When Blend Is 0.0

The tool is processed even when the input value is zero. This can be useful when this node process is scripted to trigger another task, but the value of the node is set to 0.0.

Red/Green/Blue/Alpha Channel Selector

These four buttons are used to limit the effect of the tool to specified color channels. This filter is often applied after the tool has been processed.

For example, if the Red button on a Blur tool is deselected, the blur is first applied to the image, and then the red channel from the original input will be copied back over the red channel of the result.

There are some exceptions, such as tools for which deselecting these channels causes the tool to skip processing that channel entirely. Tools that do this generally possess a set of identical RGBA buttons on the Controls tab in the tool. In this case, the buttons in the Settings and the Controls tabs are identical.

Apply Mask Inverted

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

Multiply by Mask

Selecting this option causes the RGB values of the masked image to be multiplied by the mask channel's values. This causes all pixels of the image not included in the mask (i.e., set to 0) to become black/transparent.

Use Object/Use Material (Checkboxes)

Some 3D software can render to file formats that support additional channels. Notably, the EXR file format supports Object ID and Material ID channels, which can be used as a mask for the effect. These checkboxes determine whether the channels are used, if present. The specific Material ID or Object ID affected is chosen using the next set of controls.

Correct Edges

This checkbox appears only when the Use Object or Use Material checkboxes are selected. It toggles the method used to deal with overlapping edges of objects in a multi-object image. When enabled, the Coverage and Background Color channels are used to separate and improve the effect around the edge of the object. If this option disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

For more information *see Chapter 78, "Understanding Image Channels,"* in the DaVinci Resolve Reference Manual, or Chapter 18 in the Fusion Reference Manual.

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Object ID/Material ID (Sliders)

Use these sliders to select which ID is used to create a mask from the object or material channels of an image. Use the Sample button in the same way as the Color Picker: to grab IDs from the image displayed in the view. The image or sequence must have been rendered from a 3D software package with those channels included.

Hide Incoming Connections

Enabling this checkbox can hide connection lines from incoming nodes, making a node tree appear cleaner and easier to read. When enabled, empty fields for each input on a node are displayed in the Inspector. Dragging a connected node from the node tree into the empty field hides that incoming connection line as long as the node is not selected in the node tree. When the node is selected in the node tree, the line reappears.

Comments

The Comments field is used to add notes to a tool. Click in the empty field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering.

> For more details on scripting nodes, see the Fusion scripting documentation.